Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is: A malicious actor executed a direct Denial of Service (DoS) attack using a SYN flood.

The logs show that: An unfamiliar IP address (203.0.113.0) sent an abnormally high volume of SYN packets. By log 125, the server was completely overwhelmed and stopped responding to legitimate traffic.

This event could be: A direct DoS attack, because the malicious traffic is originating from a single IP address rather than multiple sources (DDoS).

Section 2: Explain how the attack is causing the website to malfunction

Explain the three steps of the handshake:

1. The visitor sends a [SYN] packet to the web server to request a connection.
2. The web server responds with a [SYN, ACK] packet, agreeing to the connection and reserving system resources.
3. The visitor sends an [ACK] packet back to complete the connection.

Explain what happens when a malicious actor sends a large number of SYN packets all at once: The attacker sends a massive amount of SYN requests but never sends the final ACK packets. The server reserves resources for all these incomplete (half-open) connections until its capacity is completely exhausted, preventing it from handling legitimate traffic.
Explain what the logs indicate and how that affects the server: The logs indicate that the server initially tries to handle the malicious SYN requests alongside legitimate traffic. However, as the server's resources deplete from the flood, legitimate users start receiving HTTP 504 Gateway Time-out and [RST, ACK] errors, until the server completely crashes and stops responding to them.